

Product Requirements Document (PRD)

Product Name: TradeGuard – Global Trade Compliance Platform

Product URL: <https://app.tradeguard.io/>

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1. Product Overview

TradeGuard is an enterprise-grade Global Trade Compliance (GTC) platform built for supply chain, logistics, and procurement teams operating across multiple countries. It centralizes restricted-party screening, tariff classification, export/import license management, and trade documentation so that shipments never leave a warehouse — or clear customs — in violation of international trade law.

The platform integrates with ERP systems (SAP, Oracle), freight forwarders, and customs brokers to screen every order, vendor, and shipment against government watchlists in real time, before goods move.

2. Business Objectives

Primary Goals

- Reduce trade compliance violations and associated fines to near zero across all shipping lanes.
- Cut manual HS (Harmonized System) code classification time by 80% through automated, ML-assisted classification.
- Provide a single audit-ready system of record for every cross-border transaction.
- Enable trade compliance teams to clear new SKUs and new trade lanes in days, not weeks.

Secondary Goals

- Reduce customs clearance delays caused by incomplete or incorrect documentation.
- Give finance and legal teams real-time visibility into tariff/duty exposure by country and product line.

3. Target Users

Primary Users

- Trade Compliance Officers — configure screening rules, review flagged transactions, approve/deny shipments.
- Supply Chain / Logistics Coordinators — initiate shipments, attach documentation, track clearance status.

Secondary Users

- Procurement teams — screen new suppliers and vendors before onboarding.
- Finance / Legal teams — review duty exposure, sanctions risk, and compliance audit reports.
- Customs brokers (external, read-only access) — retrieve shipment documentation packets.

4. Core Features & Capabilities

4.1 Restricted & Denied Party Screening

- Real-time screening of customers, vendors, and shipment parties against global watchlists: OFAC SDN List, BIS Entity List, EU Consolidated List, UN Sanctions List, and denied-persons lists.
- Fuzzy name matching to catch spelling variants, transliterations, and aliases.
- Configurable match-confidence thresholds per watchlist, with a manual review queue for partial matches.
- Automatic re-screening of all active vendors/customers on a nightly basis as watchlists update.

4.2 HS Code Classification

- ML-assisted Harmonized System (HS) code suggestion based on product description, material composition, and prior classification history.
- Country-specific tariff schedule lookup (HTS for the US, TARIC for the EU, etc.).
- Classification confidence score with mandatory human review below a configurable threshold.
- Bulk classification for catalog imports (CSV/Excel upload of up to 10,000 SKUs per batch).

4.3 Tariff & Duty Calculation

- Landed cost calculator combining product value, freight, insurance, duty rate, and applicable trade-agreement preferences (e.g., USMCA, EU-UK TCA).
- Real-time duty rate lookup by HS code, country of origin, and destination country.
- Alerts when a shipment's origin/destination pairing changes duty exposure significantly (>15%).

4.4 Export License Management

- Tracks license requirements by product (ECCN — Export Control Classification Number), destination country, and end use.
- License application workflow with document upload, approval routing, and expiration reminders (90/60/30-day alerts).
- Blocks shipment creation if a required license is missing or expired.

4.5 Trade Documentation

- Auto-generates commercial invoices, certificates of origin, packing lists, and customs declarations from shipment data.
- Document template library, customizable per destination country's customs requirements.
- Centralized document repository with full version history, searchable by shipment ID, PO number, or vendor.

4.6 Country of Origin Determination

- Rules-of-origin engine to determine preferential trade-agreement eligibility (e.g., qualifying for USMCA duty-free treatment).
- Bill-of-materials (BOM) based origin calculation for assembled/manufactured goods.

4.7 Audit Trail & Reporting

- Immutable audit log of every screening result, classification decision, and manual override, with user, timestamp, and justification.

- Pre-built compliance reports: denied-party screening summary, duty exposure by region, license expiration forecast.
- Exportable audit packets for regulatory inspections (CSV, PDF).

5. Non-Functional Requirements

- **Data residency:** Customer shipment data must be stored in-region (US, EU, APAC) per data sovereignty requirements.
- **Availability:** 99.9% uptime SLA; screening checks must not block order processing for more than 2 seconds at p95.
- **Watchlist freshness:** Government watchlists must be refreshed within 4 hours of official publication.
- **Auditability:** All compliance decisions must be immutable and retained for a minimum of 7 years.
- **Access control:** Role-based access control (RBAC) with mandatory two-person approval for manual screening overrides.

6. Success Metrics

- Zero denied-party shipments released without a compliance override on record.
- HS classification manual review rate reduced from ~60% to under 15% within 6 months of launch.
- Average shipment clearance time reduced by 30%.
- 100% of active vendors re-screened within 24 hours of a watchlist update.

7. Assumptions & Constraints

- Assumes ERP systems expose shipment and vendor data via API or scheduled batch export.
- Watchlist data is licensed from a third-party compliance data provider and refreshed via a scheduled feed.
- Initial launch covers US, EU, and UK trade lanes; APAC lanes are planned for a future phase.
- Legal team must sign off on all screening match-confidence thresholds before go-live.